Algebra: Rationalizing Radicals

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DIRECTIONS

Rationalize the denominator. Multiply by an appropriate form of 1 (radical/radical or conjugate/conjugate). Simplify completely.

1. Rationalize the denominator: $$\frac{1}{\sqrt{3}}$$ Answer: ________________________	**2.** Rationalize the denominator: $$\frac{5}{\sqrt{7}}$$ Answer: ________________________
3. Rationalize the denominator: $$\frac{8}{\sqrt{2}}$$ Answer: ________________________	**4.** Rationalize the denominator: $$\frac{4}{\sqrt{5}}$$ Answer: ________________________
5. Rationalize the denominator: $$\frac{2+\sqrt{3}}{\sqrt{3}}$$ Answer: ________________________	**6.** Rationalize the denominator: $$\frac{6}{\sqrt{12}}$$ Answer: ________________________
7. Rationalize using the conjugate: $$\frac{3}{\sqrt{5}-2}$$ Answer: ________________________	**8.** Rationalize using the conjugate: $$\frac{6}{\sqrt{3}+1}$$ Answer: ________________________
9. Rationalize using the conjugate: $$\frac{\sqrt{2}-1}{\sqrt{2}+1}$$ Answer: ________________________	**10.** Rationalize using the conjugate: $$\frac{4}{\sqrt{6}-\sqrt{2}}$$ Answer: ________________________

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Answer Key & Solutions

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TEACHER NOTES

Items 1–6: monomial radical denominator — multiply by $\sqrt{n} / \sqrt{n}$ to clear it. Items 7–10: binomial radical denominator — multiply by the conjugate (change the sign between terms) to use the difference-of-squares pattern.

1. Rationalize the denominator: **$1 / \sqrt{3}$**

Answer: $\sqrt{3} / 3$

Multiply by $\sqrt{3}/\sqrt{3}$: $(1\cdot\sqrt{3}) / (\sqrt{3}\cdot\sqrt{3}) = \sqrt{3}/3$.

2. Rationalize the denominator: **$5 / \sqrt{7}$**

Answer: $5\sqrt{7} / 7$

Multiply by $\sqrt{7}/\sqrt{7}$: $5\sqrt{7} / 7$.

3. Rationalize the denominator: **$8 / \sqrt{2}$**

Answer: $4\sqrt{2}$

Multiply by $\sqrt{2}/\sqrt{2}$: $8\sqrt{2} / 2 = 4\sqrt{2}$.

4. Rationalize the denominator: **$4 / \sqrt{5}$**

Answer: $4\sqrt{5} / 5$

Multiply by $\sqrt{5}/\sqrt{5}$: $4\sqrt{5} / 5$.

5. Rationalize the denominator: **$(2 + \sqrt{3}) / \sqrt{3}$**

Answer: $2\sqrt{3}/3 + 1$

Multiply by $\sqrt{3}/\sqrt{3}$: $(2\sqrt{3} + 3)/3 = 2\sqrt{3}/3 + 1$.

6. Rationalize the denominator: **$6 / \sqrt{12}$**

Answer: $\sqrt{3}$

$\sqrt{12} = 2\sqrt{3}$. So $6/(2\sqrt{3}) = 3/\sqrt{3} \cdot \sqrt{3}/\sqrt{3} = 3\sqrt{3}/3 = \sqrt{3}$.

7. Rationalize using the conjugate: **$3 / (\sqrt{5} - 2)$**

Answer: $3\sqrt{5} + 6$

Conjugate: $(\sqrt{5}+2)$. Multiply: $3(\sqrt{5}+2)/(5-4) = 3(\sqrt{5}+2) = 3\sqrt{5}+6$.

8. Rationalize using the conjugate: **$6 / (\sqrt{3} + 1)$**

Answer: $3\sqrt{3} - 3$

Conjugate: $(\sqrt{3}-1)$. Multiply: $6(\sqrt{3}-1)/(3-1) = 6(\sqrt{3}-1)/2 = 3(\sqrt{3}-1)$.

9. Rationalize using the conjugate: **$(\sqrt{2} - 1) / (\sqrt{2} + 1)$**

Answer: $3 - 2\sqrt{2}$

Multiply by $(\sqrt{2}-1)/(\sqrt{2}-1)$: $(2 - 2\sqrt{2} + 1)/(2-1) = 3 - 2\sqrt{2}$.

10. Rationalize using the conjugate: **$4 / (\sqrt{6} - \sqrt{2})$**

Answer: $\sqrt{6} + \sqrt{2}$

Multiply by $(\sqrt{6}+\sqrt{2})/(\sqrt{6}+\sqrt{2})$: $4(\sqrt{6}+\sqrt{2})/(6-2) = 4(\sqrt{6}+\sqrt{2})/4 = \sqrt{6}+\sqrt{2}$.

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